

Basic Principles of Medicine 2

ER Lecture 19

Physiology of Pregnancy

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Recommended reading

- Medical Physiology - Principles for Clinical Medicine. Rhoades & Bell, 6e.
- Chapter 38 – Fertilization, and Pregnancy; pages 810-828
- <https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253092355&bookid=3211>
- Slides with blue/colored background are the text version of the explanations given in the classroom and thus, might not be showed during the lecture. However, the content MUST be read and understood.

Objectives

- SOM.MK.II.BPM2.1.ER.1.PHYS.1084. Describe the role of human chorionic gonadotropin (hCG) in the rescue of the corpus luteum in maintaining pregnancy early post-implantation
- SOM.MK.II.BPM2.1.ER.1.PHYS.1085. List the hormones secreted by the placenta and discuss the roles of sex steroids, and other hormones during of pregnancy
- SOM.MK.II.BPM2.1.ER.1.PHYS.1086. Describe the interactions between the placenta and the fetus in the pathway for production of estrogens during pregnancy.

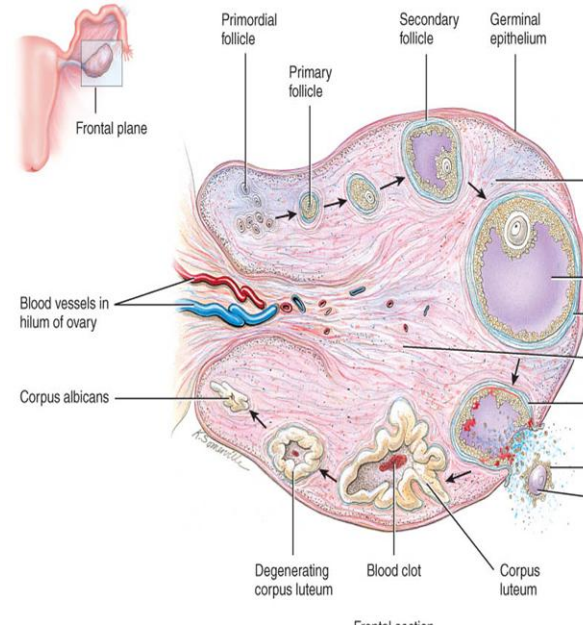
Objectives

- SOM.MK.II.BPM2.1.ER.1.PHYS.1087. Explain the maternal and fetal endocrinology and physiological changes during pregnancy.
- SOM.MK.II.BPM2.1.ER.1.PHYS.1088. List and explain two common problems associated with pregnancy (eclampsia & preeclampsia).
- SOM.MK.II.BPM2.1.ER.1.PHYS.1089. Explain how gestational diabetes can lead to problems during pregnancy.
- SOM.MK.II.BPM2.1.ER.1.PHYS.1093. Explain the basis for the inhibition of milk secretion during pregnancy and the initiation of lactation after parturition.

Human pregnancy

The big picture

- Ovulation occurs about 14th day & fertilization about 24 to 48 hours later
- hCG is secreted soon after implantation into Maternal circulation
- Implantation begins 5 to 6 days post fertilization
- Human pregnancy lasts about 280 days or 40 weeks from the last menstrual period

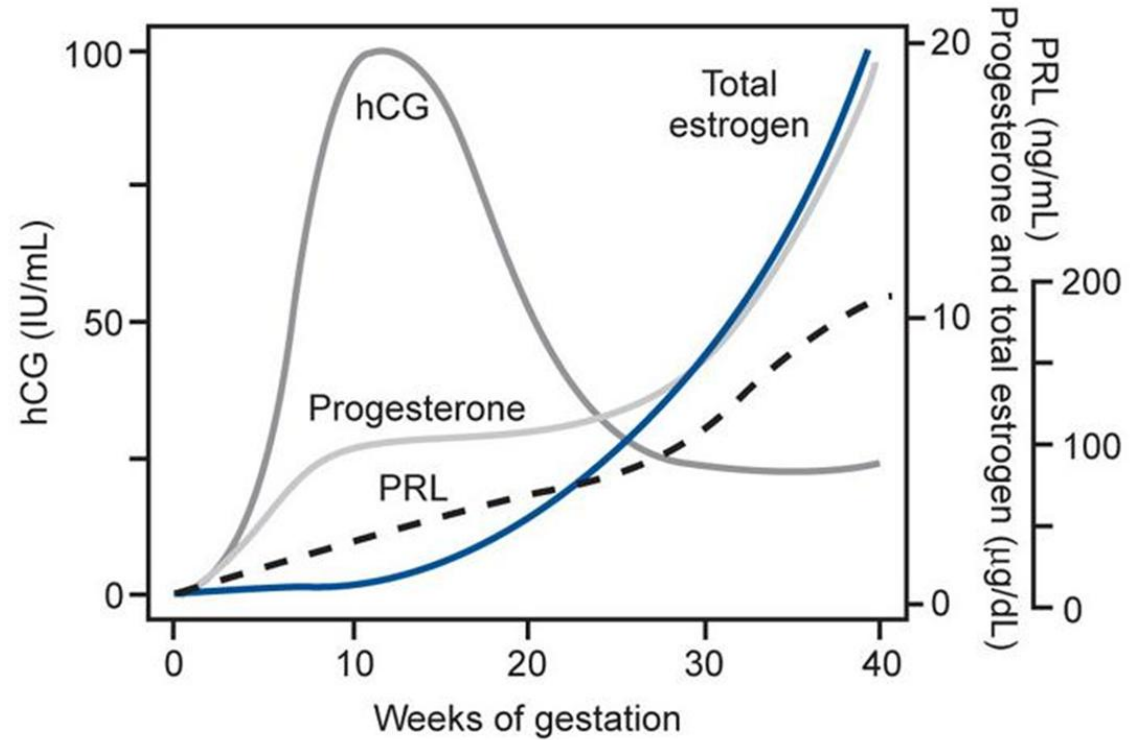


Corpus
Luteum

Hormones of Pregnancy

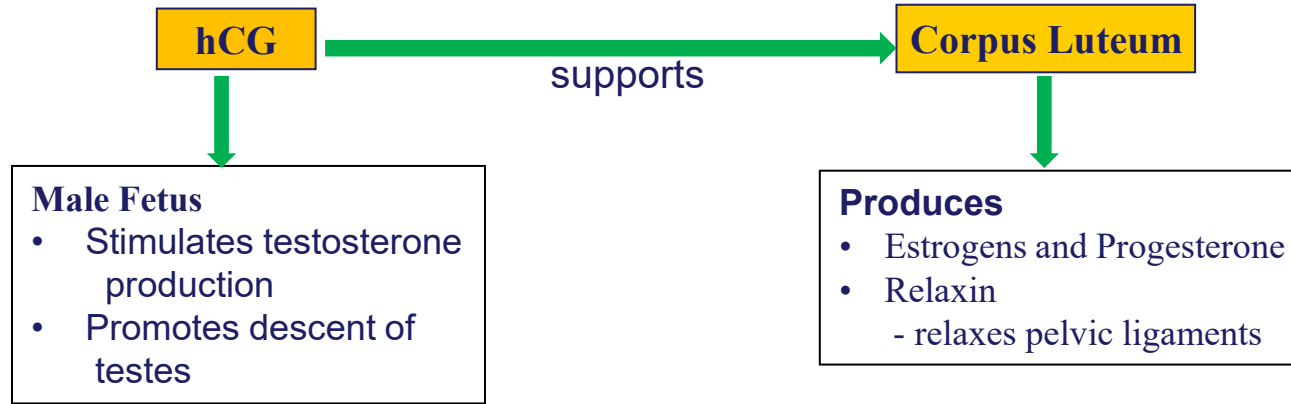
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Human chorionic gonadotropin (hCG)

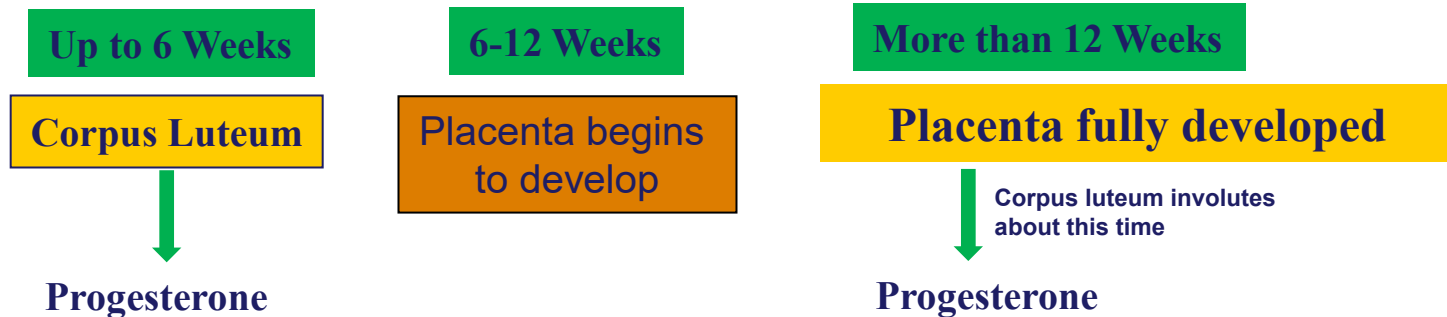


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Role of hCG



Events - in early pregnancy in mom

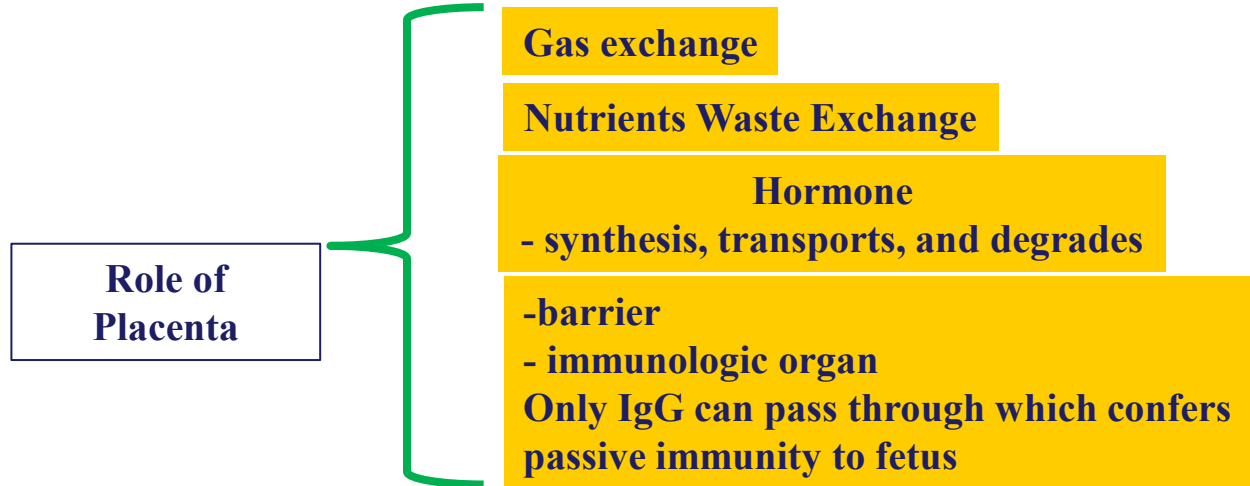


Human chorionic gonadotropin (hCG)

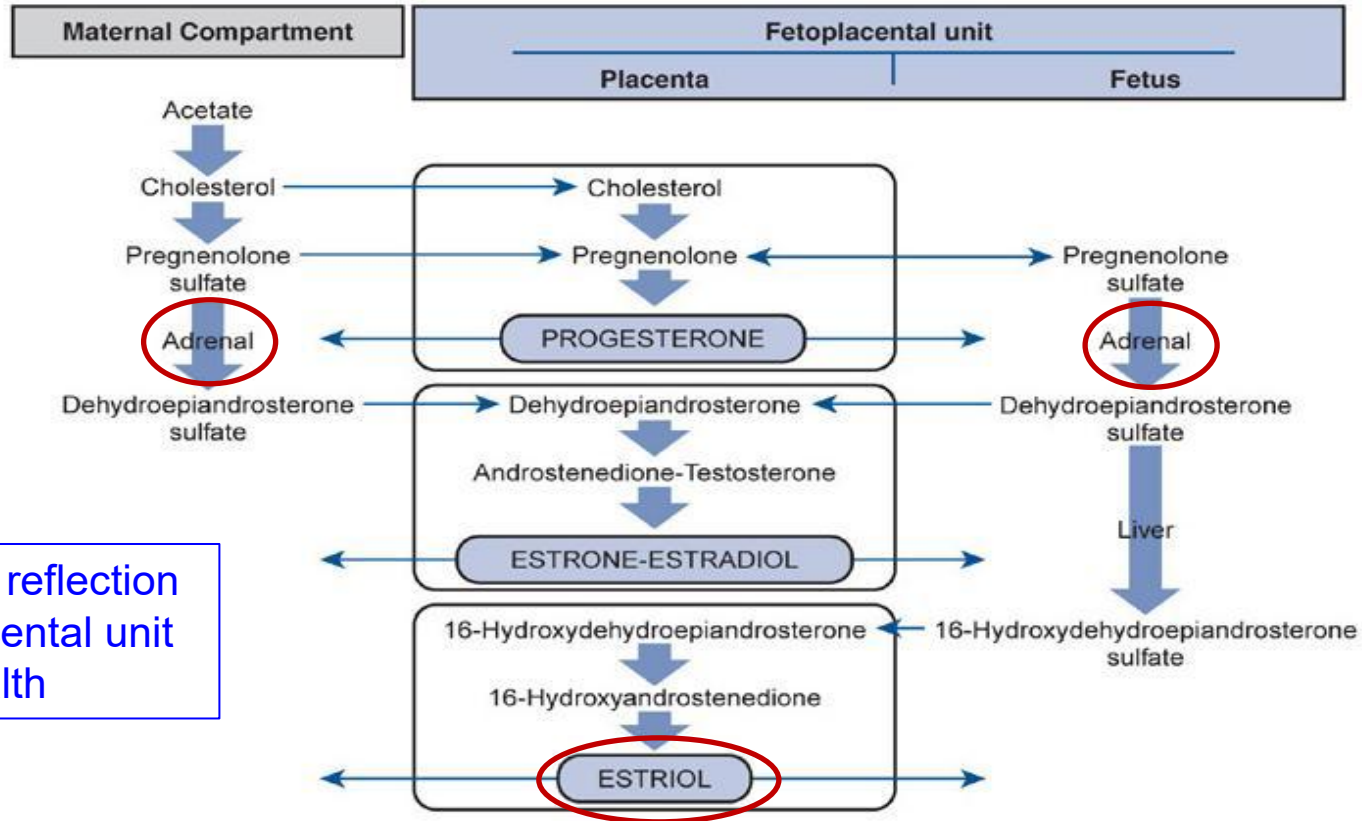
- Large glycoprotein has a long $\frac{1}{2}$ life. Secreted by syncytial cells of trophoblast into maternal circulation
- Can be measured in urine ~ 9 days after fertilization. It prevents the involution of corpus luteum
- Plasma hCG doubles every 2 days & peaks around the 13th week
- hCG has been implicated in morning sickness
- Main role of hCG is to extend the life of the corpus luteum of pregnancy and stimulation of luteal synthesis of progesterone to maintain pregnancy until placenta takes over this function.
- hCG also helps the development of the fetal adrenal in both sexes, and the testes in males.

The Placenta

- A specialized organ for exchange between maternal and fetal blood
- The fetal placenta comprises 60 to 70 **villous trees** that increase interface surface area.
- The space between fetal and maternal placenta is filled with ~500 mL of maternal blood. Blood flows in an unregulated manner at relatively high pressure (~70 mm Hg) from eroded **spiral arteries** and washes over the fetal villous trees.

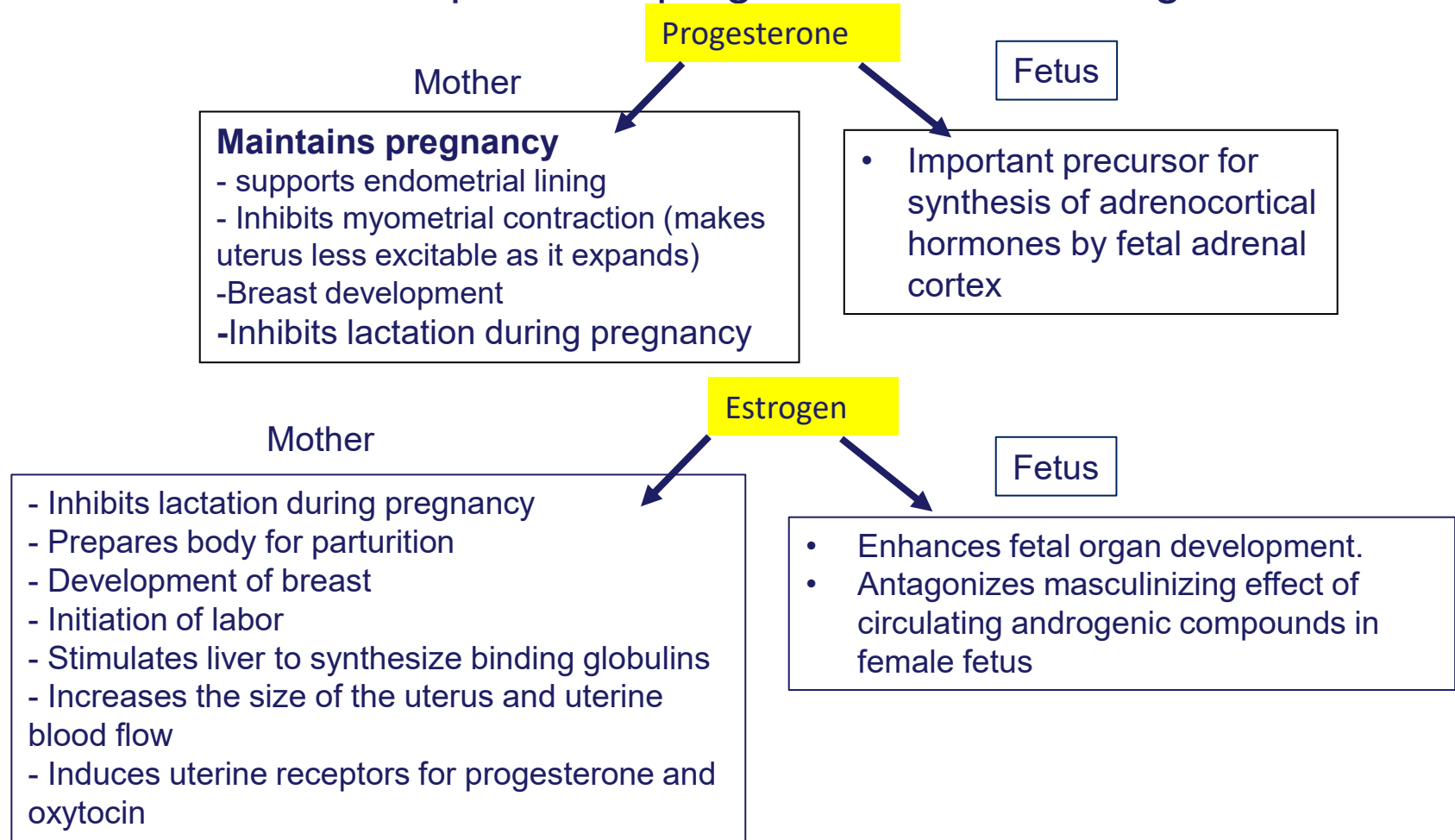


Placental – synthesis of steroid hormones

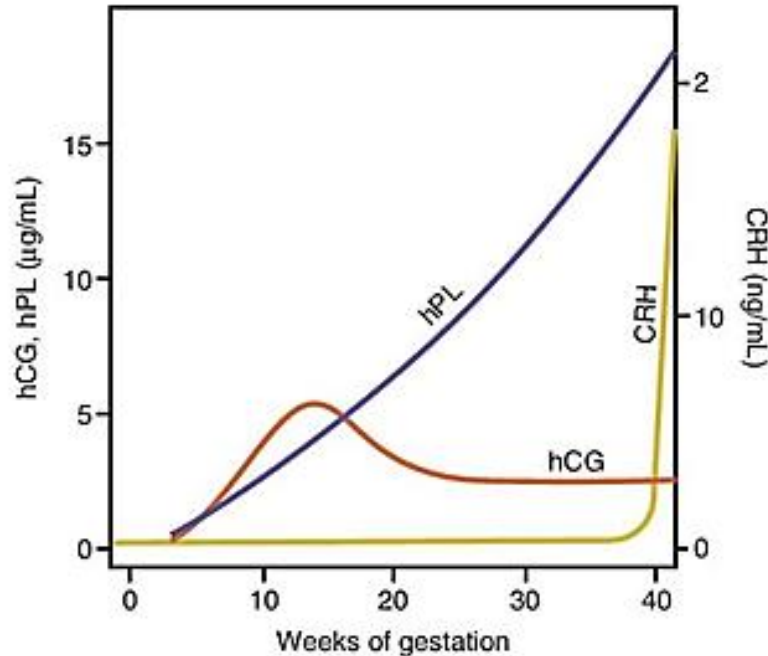


Estril is a reflection
of fetoplacental unit
health

Role of placental progesterone and estrogen



Other Hormones of Pregnancy: hPL and CRH



hPL: Human placental lactogen (it is also called human Chorionic Somatomammotropin, hCS). Amount secreted is proportional to weight of the placenta.

CRH: Corticotropin releasing hormone

Note: CRH concentration rises only near term

Other placental hormones

hPL, Human Placental Lactogen (also called human Chorionic Somatomammotropin, hCS)

- Structurally similar to Growth Hormone and Prolactin.
- Functions as maternal growth hormone of pregnancy.
- **Decreases maternal Insulin sensitivity:**
Reduces maternal glucose consumption so that more glucose is diverted to the fetus.
- Enhances fat mobilization from mother to fetus for use by the fetus and increases fetal fat stores.
- Stimulates secretions of IGF-1 and IGF-2 in fetus.

CRH

- CRH stimulates fetal pituitary to release ACTH, which increases cortisol synthesis by the fetal adrenal. Fetal cortisol increases lung maturity.
- Facilitates to initiate labor.

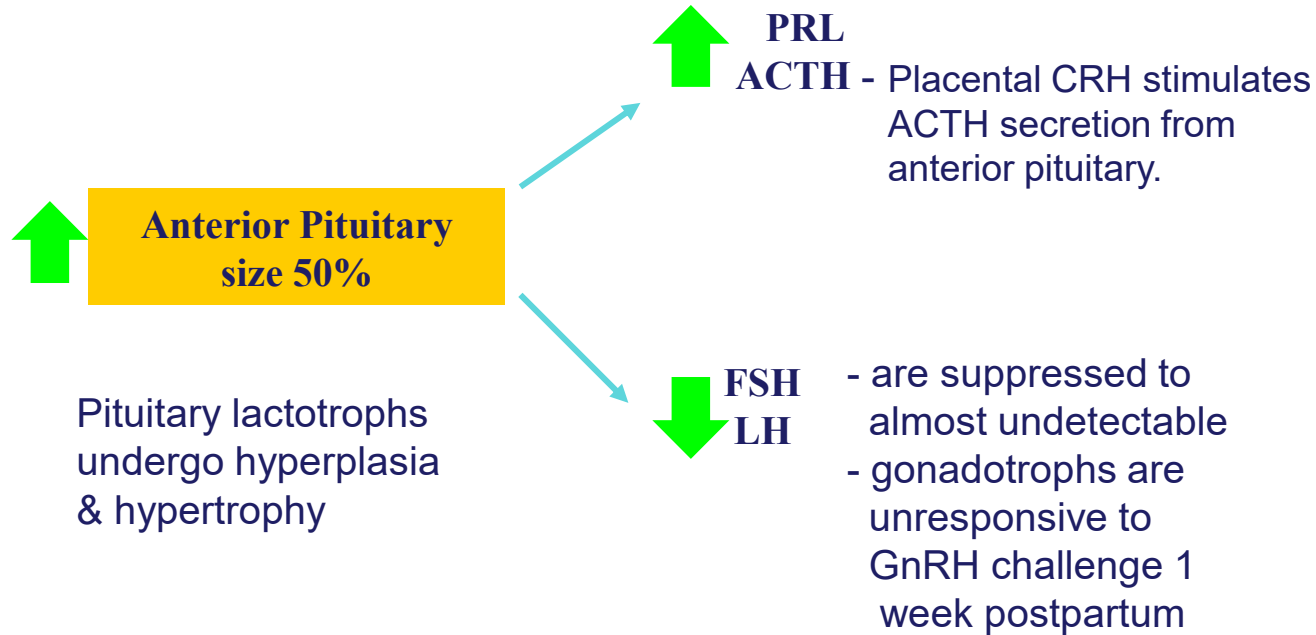
Physiological changes in mother during pregnancy

Physiological changes in mother during pregnancy

Following major physiological changes occur in the mother as Maternal Adaptations during pregnancy:

- Endocrine changes
- Cardiovascular and Hematological changes
- Pulmonary and GIT changes
- Other changes

Maternal Endocrine changes during pregnancy



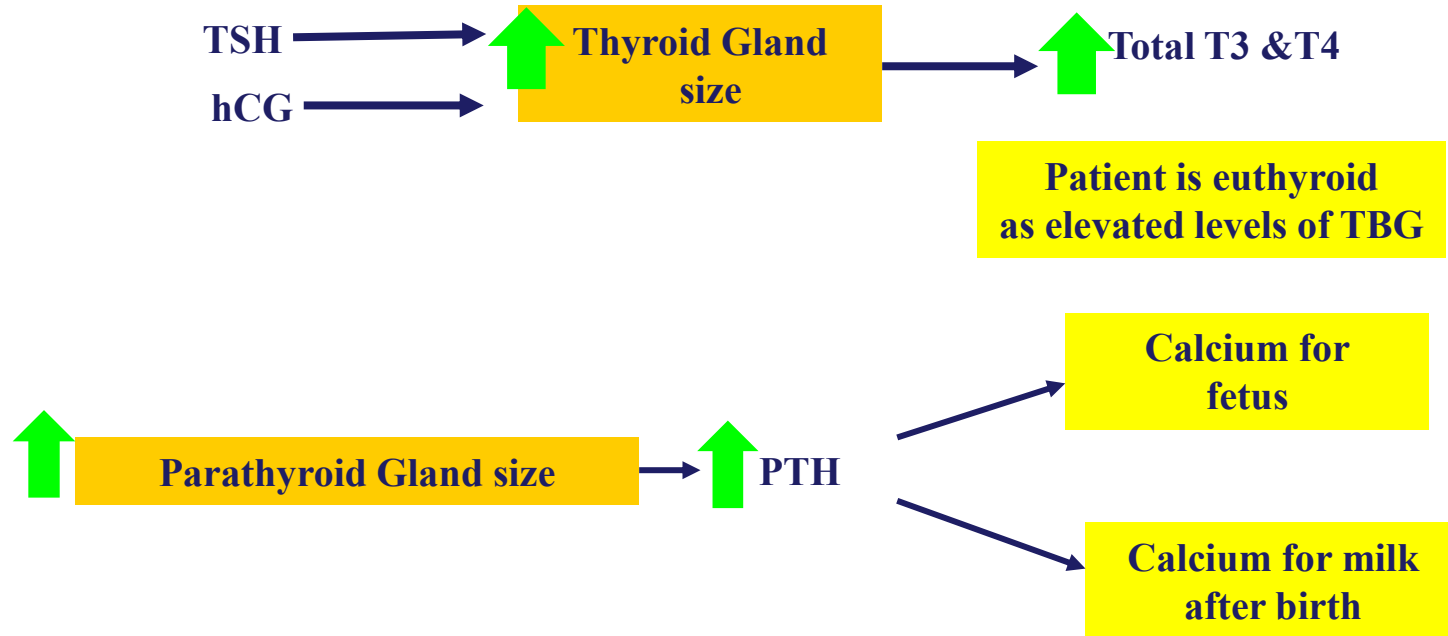
Maternal Endocrine changes during pregnancy

- High levels of estrogen & progesterone during pregnancy prevent the milk synthesis actions of prolactin on breast.
- Breast growth is stimulated by increasing progesterone, estrogen and prolactin secretion throughout pregnancy.
- Lactation is inhibited during pregnancy by progesterone produced by the placenta. **Progesterone interferes with prolactin binding to the receptors** on the alveolar cells within the breast, thereby directly suppressing milk production.

Maternal Endocrine changes during pregnancy

- High concentration of prolactin suppresses GnRH secretion from the hypothalamus
- Less production of LH and FSH results in reduced ovarian function. This is further supported by high concentrations of estrogen and progesterone from placenta.
- Temporary stoppage of menstruation and ovulation (anovulation) during pregnancy and lactation.
- The contraceptive action is transient as GnRH overcomes the suppression of prolactin after several weeks of childbirth. Not an effective method of contraception.

Maternal Endocrine changes during pregnancy



Maternal Endocrine changes during pregnancy

In response to high levels of glucose in the mother



In response to high levels of CRH & ACTH

Adrenal Gland

↑ Cortisol

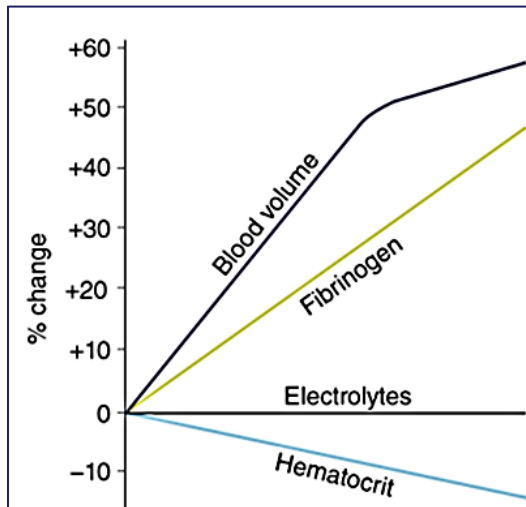
Only moderate rise in free cortisol due to ↑ CBG

- AAs for fetus
- Stretch marks
- Insulin resistance
- Weight gain & hyperphagia

Rising estrogen stimulates renin-angiotensin system and increase aldosterone concentration.

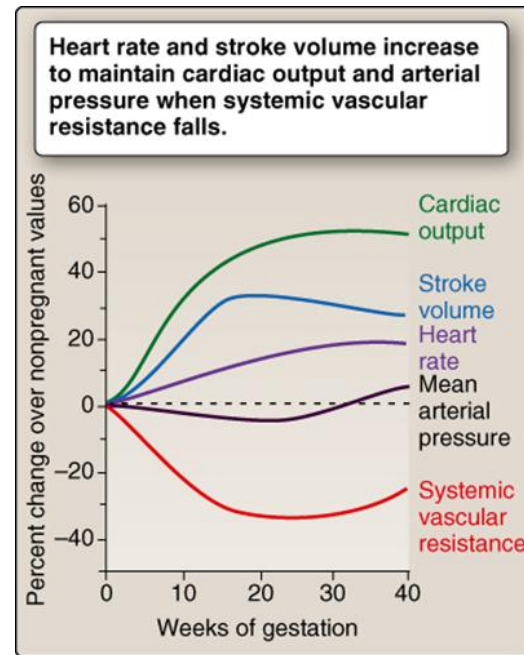
Maternal changes during pregnancy: Cardiovascular and Hematological changes

Hematological



- Blood volume ↑ by 50% in 2nd trimester
- Hematocrit ↓ slightly
- Physiologic anemia
- Clotting Factors (fibrinogen) ↑: Protects from hemorrhage during parturition

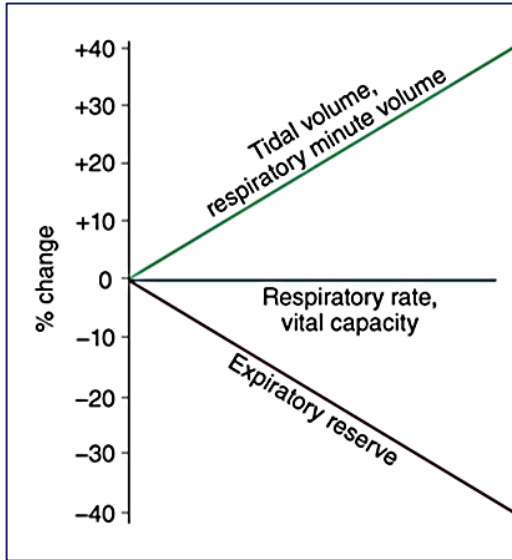
Cardiovascular



- Increase in CO is to meet with the growing demand due to several maternal changes.

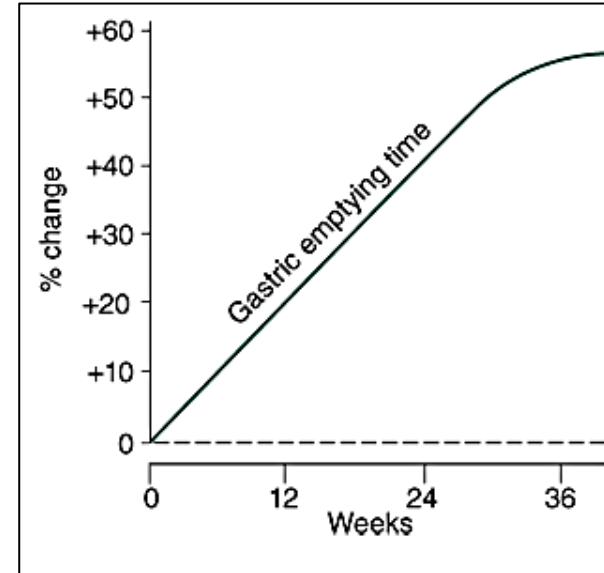
Maternal changes during pregnancy: Pulmonary and GIT changes

Pulmonary



- Increased Tidal Volume and Resp. minute volume
- Reduced Exp reserve volume
- No change in Resp rate & Vital capacity

GIT



- Reduced GI motility
- Delayed gastric emptying
- Constipation

Maternal changes during pregnancy: Cardiovascular, Hematological, Pulmonary and GIT changes

System	Parameter	Change
Cardiovascular	Heart rate	Gradually increases 20%
	Blood pressure	Gradually decreases 10% by 34 wk, then increases to prepregnancy values
	Stroke volume	Increases to maximum at 19 wk, then plateaus
	Cardiac output	Rises rapidly by 20%, then gradually increases an additional 10% by 28 wk
	Peripheral venous distention	Progressive increase to term
	Peripheral vascular resistance	Progressive decrease to term
Pulmonary	Respiratory rate	Unchanged
	Tidal volume	Increases by 30%-40%
	Expiratory reserve	Gradual decrease
	Vital capacity	Unchanged
	Respiratory minute volume	Increases by 40%

System	Parameter	Change
Blood	Volume	Increases by 50% in second trimester
	Hematocrit	Decreases slightly
	Fibrinogen	Increases
	Electrolytes	Unchanged
Gastrointestinal	Sphincter tone	Decreases
	Gastric emptying time	Increases

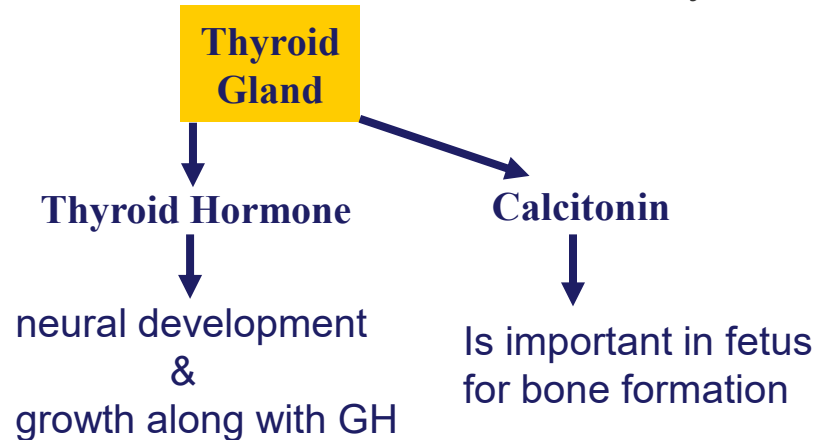
Other maternal changes during pregnancy

- Weight gain due to fat deposition, fetus & amniotic fluid
- Increase in blood volume and extravascular fluid volume
- Nausea & vomiting during 1st trimester.
- Skin pigmentation and stretch mark
- Increased GFR by 40%
- Increased nutritional requirements: Vit K, Fe, folate & Calcium

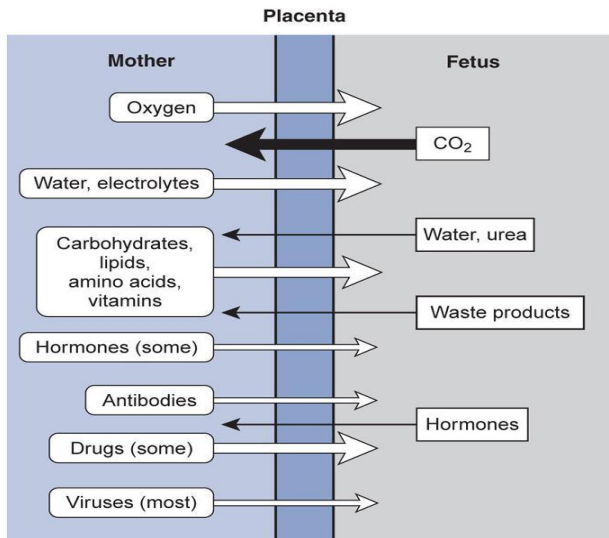
Fetal Endocrinology

Fetus

- Anterior pituitary hormones get released in circulation by ~ 20 weeks of gestation
- ACTH important for adrenal development
- **By third trimester all feedback mechanisms start working**
- ADH and oxytocin are released by **week 18**. Oxytocin may have role in parturition
- Fetal insulin, HCG influence development and growth of fetus.
- Insulin-like growth factors (IGF-1 and IGF-2) produced by most of the fetal tissues influence growth of the fetus. These factors are stimulated by hPL.



Movement of substances from mother to fetus



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	Uterine - mom		Umbilical - fetus	
	Artery	Vein	Vein	Artery
PO ₂ (mmHg)	95	40	27	15
O ₂ Hb (%sat)	98	76	68	30

Fetal Hemoglobin F vs Maternal Hemoglobin A

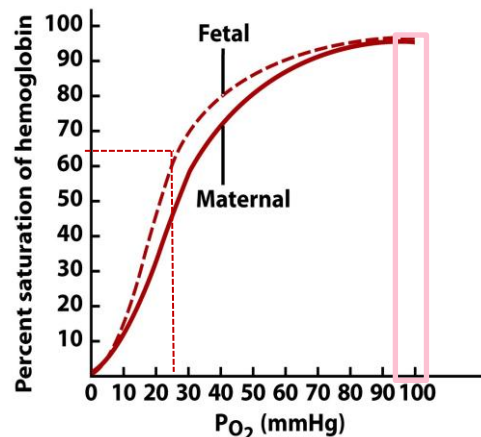


Figure 23-23 Principles of Anatomy and Physiology, 11/e
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Medical Problems Associated with Pregnancy

Preeclampsia

What causes preeclampsia?

- Caused by abnormal placental spiral arteries, endothelial dysfunction and vasoconstriction.

Clinical features

- Hypertension
 - SP > 140mmHg,
 - DP > 90mmHg
- proteinuria
 - > 300mg/day
- Edema
 - Face and hands
- Occurs mainly after 20 weeks

Eclampsia

- Clinical features: Eclampsia is characterized by all clinical features of preeclampsia, associated with generalized tonic-clonic seizures.
- Other manifestations include focal or multifocal seizures or coma.
- Death may occur due to stroke, intracranial hemorrhage or acute respiratory distress.
- Treatment: Control seizure and BP and perform immediate delivery.

Gestational Diabetes

- Gestational diabetes mellitus (GDM) is defined as abnormal glucose tolerance during pregnancy in a woman without previous diagnosis of diabetes.
- Diagnosed in the second and third trimester that was not present prior to gestation.
- Due to inadequate insulin response to match insulin resistance during pregnancy.
- Women at high risk to develop GDM are: Obesity, advanced age, family history of type 2 diabetes, and ethnicity (higher prevalence in Asian and Hispanic origins).
- Effect of GDM on fetus: Large birth size (macrosomia), preterm birth, excessive amniotic fluid (polyhydramnios) and birth injury due to the large size of the baby.
- Effect on pregnant mother: Hypertension, preeclampsia and possibility of developing type2 diabetes mellitus in future.

Key concepts

Pregnancy begins with fertilization. The placenta exchanges nutrients, hormones, and waste products between fetal and maternal circulations. The placenta secretes **human chorionic gonadotropin**, **progesterone**, and **estrogens**.

- Supporting the needs of a developing fetus involves most of a mother's organ systems, including the **CVS**, **kidneys** (increased disposal of waste products), **lungs** (~30% increase in O_2 demand), **GI tract**, **liver**, and **skin** (thermoregulation).
- Maternal **cardiac output** increases ~50% during gestation, accomplished through increases in **heart rate** and **stroke volume (SV)**. SV increases as a result of fluid retention and increased **circulating blood volume**.
- Blood volume increases faster than red blood cell production, causing a **physiologic anemia of pregnancy**. The resulting decrease in blood **viscosity** reduces **shear stress** on the heart and vasculature lining.
- The uterus and its contents gain significant weight during pregnancy, which, depending on posture, compresses and impairs blood flow from the mother's lower extremities. The result is **pedal edema**.
- **Fetal hemoglobin (HbF)** has a high O_2 affinity, helps compensate for the fact that the placenta is a less efficient route for O_2 transfer than the lungs and allows fetal blood to carry near-adult levels of O_2 .

References

- Medical Physiology – Principles for Clinical medicine – Rodney A Rhoades & David A. Bell – 6th edition
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- Patel, Bansari, et al. "The Endocrinology of Pregnancy." *Greenspan's Basic & Clinical Endocrinology, 10e* Eds. David G. Gardner, and Dolores Shoback. McGraw Hill.
- "Maternal Physiology." *Williams Obstetrics*, F. Gary Cunningham, et al. McGraw Hill.
- Lippincott® Illustrated Reviews: Physiology.